

SCAMMON BAY, AK, USA

WMO: 703054

Lat: 61.845N

Lon: 165.571W

Elev: 4

StdP: 101.27

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 00108

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.8	-26.3	-32.2	0.2	-28.8	-28.8	0.3	-25.8	15.6	-8.6	14.5	-6.3	1.1	100	0.874

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.7	18.8	14.9	17.2	13.8	15.6	13.1	15.3	18.0	14.2	16.4	13.3	15.2	4.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	10.0	16.4	13.0	9.3	15.0	12.3	8.9	14.5	43.1	17.9	39.9	16.5	37.5	15.2	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	12.1	10.7	DB	-28.0	23.2	2.7	3.0	-29.9	25.4	-31.5	27.1	-33.0	28.8	-34.9	30.9	(4)
			WB	-27.9	17.2	2.6	2.4	-29.8	18.9	-31.3	20.4	-32.8	21.7	-34.7	23.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.3	-14.0	-10.8	-12.6	-5.9	2.5	9.1	11.6	11.3	7.9	1.6	-6.6	-10.1
	DBStd	10.80	8.82	7.87	6.49	5.55	4.00	3.46	2.72	2.43	2.85	3.74	5.87	6.28
	HDD10.0	4275	743	583	701	478	234	56	12	13	72	261	499	622
	HDD18.3	7162	1002	817	959	728	490	278	209	218	313	519	749	880
	CDD10.0	157	0	0	0	0	2	29	63	54	9	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	4	0	0	0	0	0	0	4	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.2	3.2	4.6	4.1	4.2	4.1	4.7	4.5	4.3	4.4	4.4	4.3	3.4
Precipitation	PrecAvg	714	49	48	34	30	38	54	77	98	96	71	66	53
	PrecMax	974	110	117	107	60	82	82	117	147	170	142	130	111
	PrecMin	573	10	14	4	7	4	24	46	34	14	20	13	12
	PrecStd	94	24	26	22	13	18	15	21	26	29	25	30	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	2.4	2.2	6.0	16.3	20.6	22.9	20.8	16.0	10.5	5.8	4.0
		MCWB	2.6	1.4	1.3	4.0	11.9	14.9	17.7	15.8	12.7	8.5	4.5	3.5
	2%	DB	1.9	1.3	0.9	3.6	12.4	17.6	19.1	17.9	13.8	9.1	3.6	1.9
		MCWB	1.0	0.7	0.3	2.3	9.3	13.4	15.4	14.2	11.8	7.7	2.7	1.3
	5%	DB	0.8	0.9	-0.5	2.3	9.9	15.5	17.4	16.3	12.6	8.1	2.4	0.9
		MCWB	0.3	0.3	-1.0	1.3	7.9	12.5	14.1	13.9	11.0	6.9	1.7	0.4
	10%	DB	-1.2	-0.3	-2.4	1.2	7.8	13.9	15.9	14.7	11.8	7.0	1.1	-0.9
		MCWB	-1.7	-0.7	-3.0	0.5	6.0	11.4	13.3	13.0	10.5	5.9	0.4	-1.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	1.7	1.4	4.2	11.6	15.6	18.8	16.2	13.6	9.1	4.9	3.5
		MCDB	3.7	1.9	1.8	5.8	14.8	19.2	22.4	18.9	14.4	9.8	5.5	3.8
	2%	WB	1.3	0.8	0.3	2.4	9.5	14.0	15.9	15.0	12.3	8.0	2.9	1.4
		MCDB	1.9	1.1	0.8	3.4	11.6	16.9	18.6	17.2	13.2	8.8	3.4	1.6
	5%	WB	0.2	0.1	-1.0	1.3	7.7	12.8	14.4	14.1	11.3	7.1	1.7	0.4
		MCDB	0.6	0.6	-0.6	2.1	9.8	15.1	16.8	16.0	12.2	8.0	2.3	0.8
	10%	WB	-1.8	-0.9	-3.0	0.4	6.2	11.7	13.4	13.2	10.4	6.0	0.4	-1.1
		MCDB	-1.4	-0.4	-2.4	1.1	7.8	13.7	15.5	14.8	11.4	7.0	1.0	-0.7
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.2	6.5	5.6	4.8	5.1	4.7	4.3	3.9	3.5	4.9	5.4
		MCDBR	5.8	5.8	6.3	5.8	8.4	7.9	7.7	6.7	5.4	3.9	4.7	5.1
		MCWBR	5.5	5.5	5.6	4.9	5.7	4.9	4.8	4.1	3.7	3.5	4.8	4.9
	5% WB	MCDBR	5.8	5.8	5.5	5.6	8.0	7.3	7.2	5.8	4.6	3.9	4.5	5.2
		MCWBR	5.6	5.6	5.3	4.8	5.8	4.9	4.7	3.8	3.6	3.6	4.6	5.0
Clear-Sky Solar Irradiance	taub	0.217	0.282	0.288	0.355	0.375	0.373	0.382	0.400	0.340	0.302	0.250	0.210	
	taud	2.033	2.121	2.192	2.042	2.201	2.393	2.406	2.301	2.465	2.353	2.022	1.880	
	Ebn at Noon	483	670	824	820	839	853	833	776	772	657	402	263	
	Edh at Noon	36	70	93	134	125	106	102	105	74	56	37	24	
All-Sky Solar Radiation	RadAvg	0.30	1.03	2.79	4.50	5.09	5.18	4.17	3.32	2.19	1.09	0.34	0.14	
	RadStd	0.04	0.12	0.31	0.33	0.51	0.50	0.29	0.46	0.19	0.09	0.05	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page